

Software Requirements Specification

A Mobile Application for Facility Management

Team[Tech Nest]

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Tutorial Details

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1. Introduction

The purpose of this mobile application is to improve the university's facilities such as: facility cleanliness. The intended audience are the GIU committee (Users), facility management, Admin and workers. The app creates direct communication between users and facility management to ensure that complaints are heard, tracked and resolved.

1.1 Product Vision & Scope

The aim is to create a mobile application that is a fast, reliable and transparent maintenance communication platform that ensures campus facilities are maintained efficiently.

The goals are:

- Reduce maintenance response time.
- Improve accuracy of issue reporting through photo evidence.
- Provide real-time tracking of maintenance tasks.
- Enable automated alerts for restocking tissues, soap, and baskets.
- Offer a user-friendly interface where tickets can be submitted in under one minute.

1.2 Definitions and Acronyms

SRS: Software Requirements Specification

Ticket: A digital record representing a maintenance issue

RBAC: Role-Based Access Control

Facility Management (FM): Department responsible for campus maintenance

Worker: Maintenance staff assigned to resolve tickets

Admin: System administrator managing users and system configuration

2. Functional Requirements

Describes what the system should do, first in the form of user stories then as a list of functional requirements for each user/role. A UML use-case diagram should then be included to link specific features and requirements.

Community Member:

FR-CM-01 (Register an account) - High

The system shall allow the user to register a new account. by providing required information (e.g., name, email, and password).

FR-CM-02 (Login and Create Ticket): - High

The system shall allow Community Members to log in using a valid email and password & create a new maintenance ticket.

FR-CM-03 (Upload Image): - High

The system shall allow a logged-in user to access photos and camera

The system shall allow a logged-in user upload photos from gallery

The system shall allow a logged-in user capture photo

FR-CM-04 (Building Selection): -High

The system shall allow the user to choose the building where the issue is located.

FR-CM-05 (Floor Selection): -High

The system shall allow the user to select the floor number where the issue is located.

FR-CM-06 (Nearest Room Number Entry):- Medium

The system shall allow the user to enter the nearest room number adjacent to the bathroom to identify the exact location.

FR-CM-07 (Issue Category Selection):- High

The system shall allow the user to select the issues categories (Tissue Restocking, Soap Refill, Trash Overflow, Door Malfunction, Plumbing Issue, Other)

FR-CM-08 (Description Field): -Medium

The system shall allow the user to enter additional details about the issue.

FR- CM-09 (Notification – Submission): -Medium

The system shall notify the user that the ticket has been successfully submitted.

FR-CM-10 (Notification – Resolution):- Low

The system shall notify the user when the ticket status(pending, in-progress, resolved) is marked as resolved.

FR-CM-11 (View Ticket Status): -Low

The system shall allow the user to track the status of submitted tickets.

FR-CM-12 (Change password.): -Medium

The system shall allow the user to change their password.

FR-CM-13: (Log Out)

The system shall allow a Community Member to log out of the application.

Facility Management:

FR-FM-01: Account Creation

The system shall allow a Facility Manager to create an account.

FR-FM-02: User Authentication

The system shall allow a Facility Manager to log in using a valid email and password.

FR-FM-03: Logout

The system shall allow a Facility Manager to log out of the application.

FR-FM-04 (Receive Ticket Alerts): -High

The system shall notify the facility manager when a new ticket is submitted

FR-FM-05 (View Tickets Details): -Medium

The system shall allow the facility manager to view ticket details such as building, floor, nearest room number, issue category, uploaded image and description.

FR-FM-06 (Assign Worker): -High

The system shall allow the user to assign tickets to specific workers

FR-FM-07 (Update Ticket Status): -High

The system shall allow the user to change the ticket status to: pending, assigned and resolved

FR-FM-08 (Filter Tickets): -Medium

As a facility management member, I should be able to filter all the tickets based on their tag/location to make it easier to search for tasks.

FR-FM-09(View profiles) -Medium

The system shall allow the user to view workers' profiles.

FR-FM-10 (Merge similar requests) - Medium

The system shall allow the user to merge similar requests submitted

FR- FM-11 (Add priority) - High

The system shall allow facility management members to assign a priority level (high, medium, low) when creating or updating a task.

FR- FM-12 (Close Issue) - High

The system shall allow a Facility Manager to close an issue once it has been resolved.

Worker:**FR-W-01: Account Creation**

The system shall allow a Worker to create an account.

FR-W-02: User Authentication

The system shall allow a Worker to log in using a valid email and password.

FR-03: Logout

The system shall allow a Worker to log out of the application.

FR-04: View Assigned Tasks: -High

The system shall show workers only the tickets assigned to them.⁴

FR-W-05: Comment on Issue

The system shall allow a Worker to add comments to an issue.

FR-W-06: Update Issue Status to In Progress

The system shall allow a Worker to mark an assigned issue as **In Progress**.

FR-W-07 : Precise Location Display: - High

The system shall show workers to see the exact location including: building, floor and nearest room number.

FR-W-08 Update Progress: - High

The system shall allow workers to update ticket status to “Assigned” and “Resolved”.

FR-W-09 (Upload Resolution Image): - High

The System shall allow the user to upload a photo as proof of issue resolution.

Admin:**FR-A-01: Admin Authentication**

The system shall allow a System Admin to log in.

FR-A-02: Logout

The system shall allow a System Admin to log out.

FR-A-03 (User Management): -High

The system shall allow administrators to create, edit, suspend, or delete user accounts.

FR-A-04 (Role Assignment): - High

The system allows the enforcement of Role-Based Access Control (RBAC) and the assignment of roles (Community Member, Facility Management, Worker).

The system allows the creation of credentials for the Facility Management Worker.

FR-A-05 (System Monitoring): - High

The system allows users to monitor ticket statistics and system activity. (Dashboard)

FR-A-06 (Reset password) - medium

The system shall allow the user to reset their password.

FR-A-07: View Registered Users

The system shall allow a System Admin to view all registered users in the system.

2.1 User stories:

A GIU community member user story:

- A student notices that tissues are finished in the women's bathroom. She opens the app, takes a photo, selects the building and floor, and enters the nearest room number (e.g., M.204). She adds a short description and submits the ticket. She receives a notification confirming submission and another notification when the issue is resolved.

User 1: Community member

US-CM-01: As a community member, I can register an account and login to be able to use the app using email and password.

US-CM-02: As a community member, I am allowed to create a new maintenance ticket.

US-CM-03: As a community member, I am allowed to upload at least one image per ticket.

US-CM-04: As a community member, I can select the building where the issue is located.

US-CM-05: As a community member, I can select the floor number where the issue is located.

US-CM-06: As a community member, I can enter the nearest room number adjacent to the bathroom to identify the exact location.

US-CM-07: As a community member, I can select an issue category: Tissue Restocking, Soap Refill, Trash Overflow, Door Malfunction, Plumbing Issue and Other

US-CM-08: As a community member, I can enter additional details about the issue.

US-CM-09: As a community member, I can be notified once the ticket has been successfully submitted.

US-CM-10: As a community member, I can be notified when the ticket status is marked as resolved.

US-CM-11: As a community member, I can track the status of submitted tickets by viewing it.

US-CM-12: As a community member, I can change the password.

US-CM-13: As a Community Member, I should be able to log out so that my data remains secure on shared devices.

User 2: Facility Management

US-FM-01: As a Facility Manager, I should be able to create an account so that I can log in to the application.

US-FM-02 : As a Facility Manager, I should be able to log in using my email and password so that I can access the application.

US-FM-03 : As a Facility Manager, I should be able to log out so that my data remains secure on shared devices

US-FM-04 : As a facility manager, I can be notified when a new ticket is submitted.

US-FM-05 : As a facility manager, I can view: building, floor, nearest room number, issue category, uploaded image and description

US-FM-06 : As a facility manager, I can assign tickets to specific workers.

US-FM-07: As a facility manager, I can change the ticket status to: pending, assigned and Resolved

US-FM-08 : As a facility manager, I can filter tickets by building, floor, category, or status.

US-FM-09 : As a facility manager, I can view workers' profiles.

US-FM-10 : As a facility manager, I can merge similar requests submitted.

US-FM-11 : As a facility manager, I should be able to set a priority level to a task so that urgent tasks are handled first.

User 3: Worker

US-W-01: As a Worker, I should be able to create an account so that I can log in to the application.

US-W-02: As a Worker, I should be able to log in using my email and password so that I can access the application

US-W-03: As a Worker, I should be able to log out so that my data remains secure on shared devices.

US-W-04: As a worker, I can only view tickets assigned to me.

US-W-05: As a worker, I can see the exact location including: building, floor and nearest room number

US-W-06: As a Worker, I should be able to comment on an issue to indicate that the work is completed so that the facility manager can update its status.

US-W-07: As a Worker, I should be able to mark an issue as “In Progress” so that the facility manager and community member know that it is being worked on.

US-W-08: As a worker, I can upload a photo as proof of issue resolution.

User 4: Admin

US-A-01: As a System Admin, I should be able to log in so that I can access the application.

US-A-02 : As a System Admin, I should be able to log out so that my data remains secure on shared devices.

US-A-03: As a system admin, I can create, suspend or delete user accounts.

US-A-04: As system admin, I can enforce Role-Based Access Control (RBAC) and assign roles (Community Member, Facility Management, Worker). Admin creates credentials for the Facility Management Worker.

US-A-05: As a system admin, I can monitor ticket statistics and system activity. (Dashboard)

US-A-06: As a system admin, I can reset passwords.

2.2 UML Use-case Diagram:

<https://drive.google.com/file/d/12A5S9xHF6mlahFc3tPCpQ2scGIfV9KVF/view?usp=sharing>

3. Non-functional Requirements:

Details the performance, usability, security, and other constraints of the system.

These items can be re-used:

- Response Time: Page load times under three seconds
- Concurrent Users: Support for at least 500 simultaneous users
- Image Upload Speed: Images should upload within 5 seconds on stable networks.
- API Security: Rate limiting and input validation
- Backup & Recovery: Automated daily backups with disaster recovery plan
- Usability: Intuitive UI such that tickets for issues can be submitted in under one minute.
- Reliability: Must handle image uploads robustly, even on unstable network connections.

4. Constraints:

4.1 Technical Constraints :

- Internet connectivity is required for all system operations (issue submission, updates, etc.).
- Image uploads (issue photos) must be limited in size (e.g., max 5MB) to maintain performance

4.2 Security Constraints:

- Secure login (email/ID & password)
- User data must be protected using encryption (HTTPS).
- Role-based access control must be enforced (Community Member, Worker, Facility Manager, Admin roles).

4.3 Operational Constraints

- Daily automated backups must be performed.
- The system depends on users to provide accurate issue data (location, description, photos).

4.4 Design Constraints

- The UI must be simple and user-friendly for non-technical users.
- The system should support clear status tracking (Pending, In Progress, Resolved).

5. ERD:

https://drive.google.com/file/d/1er3mS3lHpX2PxW-LfY6QMcrZg_ksJ8uo/view?usp=sharing

6. Technology Stack:

Frontend : React Native
Backend: node.js
Database : Postgress

7. Future Scope:

7.1 GPS & Map Integration:
Location services can automatically detect the user’s location
Display reported issues on a map for better tracking and visualization.

7.2 Rating & Feedback System

Allow Community Members to rate and give feedback on completed issues, improving accountability and service quality.

7.3 Multi-Language Support

Support multiple languages (e.g., English and Arabic) to make the system more accessible to a wider range of users

8. Product BackLog

BackLog ID	UserStoryID	Priority
Community Member		
PB-CM-01	US-CM-01	High
PB-CM-02	US-CM-02	High
PB-CM-03	US-CM-03	High
PB-CM-04	US-CM-04	High
PB-CM-05	US-CM-05	High
PB-CM-06	US-CM-06	High
PB-CM-07	US-CM-07	High
PB-CM-08	US-CM-08	Medium
PB-CM-09	US-CM-09	Medium
PB-CM-10	US-CM-10	Low
PB-CM-11	US-CM-11	Low
PB-CM-12	US-CM-12	Low
PB-CM-13	US-CM-13	Medium

Facility Management		
PB-FM-01	US-FM-01	High
PB-FM-02	US-FM-02	High
PB-FM-03	US-FM-03	High
PB-FM-04	US-FM-04	High
PB-FM-05	US-FM-05	High
PB-FM-06	US-FM-06	High
PB-FM-07	US-FM-07	High
PB-FM-08	US-FM-08	Medium
PB-FM-09	US-FM-09	Low
PB-FM-10	US-FM-10	Low
PB-FM-11	US-FM-11	Medium

Worker		
PB-W-01	US-W-01	High
PB-W-02	US-W-02	High
PB-W-03	US-W-03	High
PB-W-04	US-W-04	High
PB-W-05	US-W-05	High
PB-W-06	US-W-06	Medium
PB-W-07	US-W-07	High
PB-W-08	US-W-08	Medium

Admin		
PB-A-01	US-A-01	High
PB-A-02	US-A-02	High
PB-A-03	US-A-03	High
PB-A-04	US-A-04	High
PB-A-05	US-A-05	Medium
PB-A-06	US-A-06	Medium